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Assessment of factors influencing farmers' adoption of flood-tolerant rice varieties in Nepal

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ABSTRACT

The impact of climate change has become serious and the agricultural sector is especially vulnerable. Adopting stress-tolerant varieties is one of the adaptive strategies, but the adoption of such varieties remains low. Varietal maintenance research, which adds biotic or abiotic stress tolerance to existing varieties, can reduce the climate vulnerability for farmers. In this context, factors that could accelerate the adoption of submergence-tolerant rice (SUB1) varieties are assessed in Chitwan district of Nepal, where rice crops are increasingly exposed to submergence conditions due to flooding. Household surveys were conducted in October 2019 and 2022 in two regions – Madi and Bharatpur – of the district. Data from 268 households were analyzed using a pooled binomial logit model, which can efficiently identify farmers' adoption decisions. Results reveal that peer learning and an expanding farmer network accelerate the adoption rate. Farmers' adoption decision also depends on their access to inputs, particularly pesticides. Policies to promote climate resilience should also focus on awareness of SUB1 variety, as well as reducing the constraints faced by resource-poor farmers. Understanding farmers' decisions in local contexts can promote adoption of stress-tolerant varieties.

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1. Introduction

The economic impact of climate change has worsened as extreme weather events become more common (CRED & UNDRR, 2020). According to the Intergovernmental Panel on Climate Change (IPCC), populations in developing countries and with limited economic opportunities are especially vulnerable (IPCC, 2023). A simulation of climate change's impacts through 2030 suggests that they are most serious in sub-Saharan Africa and South Asia, likely pushing them into extreme poverty (Jafino et al., 2020).

Studies have reported adverse impacts of climate change on farmer livelihoods and farm production (Dahal et al., 2023; Gauchan et al., 2012; Malla, 2009; Pandey et al., 2012). Nepal is one of those countries struggling with climate risks such as floods and droughts, especially in the Terai region, known as Nepal's food bowl. Frequent flooding due to monsoonal rains has become common (Mirza, 2011), affecting households and their livelihoods (UNDRR, 2019). Agriculture is one of Nepal's principal economic sectors, contributing 25.5% to its GDP in fiscal year 2020/21, and many households depend on agriculture for their livelihood (Ministry of Finance, 2022). Alleviating these risks is therefore an urgent task (Anderson et al., 2020) and several potential adaptation measures, specifically in water-logging situations, are available (Alam et al., 2017). One of them is to adopt new varieties such as those with stress tolerance (Hussain et al., 2020; Zhou et al., 2018).

The maintenance research in breeding science is to add biotic or abiotic stress-tolerance characteristics to widely adopted high-yield varieties (HYVs) (Bohn & Byerlee, 1993; Byerlee, 1996; Marasas et al., 2003). Though not explicitly known as maintenance research, adding stress tolerance to existing varieties is now common in rice breeding (Pixley et al., 2023), including tolerance to heat (Visakh et al., 2024), to salinity (Mheni et al., 2024), and to both stagnant water and submergence conditions (Kato et al., 2019).

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Despite these new STVs, however, their diffusion has been relatively slow (Bairagi et al., 2021; Mehar et al., 2022; Veettil et al., 2021; Vergara et al., 2023; Yamano et al., 2018). Many studies have focused on HYV adoption, and a farmer's learning process is found to have been among the most critical factors (Conley & Udry, 2010; Foster & Rosenzweig, 1995). In particular, farmers can learn the productivity and profitability advantages of HYVs *every season*. However, this may not be true for STVs, where learning is necessarily stochastic and not apparent to farmers in every season (Lybbert & Bell, 2010; Yorobe et al., 2016). STVs adoption process thus may differ from that of HYVs.

Given that benefits of STVs are stochastic and not always observable, several studies have pointed to the importance of learning opportunities. Yamano et al. (2018) find in Bangladesh that crop submergence in the previous year boosted the adoption of the flood-tolerant variety. The adoption also tends to be greater if early adopters are living nearby. Bairagi et al. (2021) find that farmer organizations, training programs, and access to information on new seed types via neighbours are the main drivers of STVs adoption. Field trials and training activities are also among effective approaches to boost farmers' adoption of STVs in Cambodia (Vergara et al., 2023), where the main source of knowledge is farmer-to-farmer exchange. Education and seed accessibility are also seen to be of importance in India (Mehar et al., 2022).

In STV adoption analysis, it would be ideal to compare one with its original HYV. A study of Emerick et al. (2016) employed randomized control trials (RCTs) comparing an STV with its original, non-tolerant variety, examining the effect of adding flood-tolerance characteristics to the farmer's field. In Nepal, HYVs of rice have been widely adopted (Gauchan et al., 2012; Ghimire & Huang, 2016; Ghimire et al., 2015; Joshi & Bauer, 2006; Upadhyaya & Thapa, 1994). However, the original variety of STV, in our case Sambha Mansuli, was no longer planted in Terai. The original variety was widely replaced by a different HYV, Sabitri. The scientific comparison between the new STV and its original variety, as studied by Emerick et al. (2016), is no longer available or feasible in our study site. Though STVs that bring additional advantages such as submergence or drought tolerance to HYVs have been increasingly available to farmers (Dixit et al., 2017), an important question is to understand the adoption of an STVs in their current and local context. STVs are known to reduce the vulnerability of Nepalese agriculture to climate change, but the reasons for their slow adoption are not yet known. The purpose of the present study is to determine the factors that may accelerate adoption of STVs in Nepal. This requires knowledge about the current variety farmers use and the corresponding household characteristics.

STV is also expected to reduce the economic disparity by reducing productivity gap. It is known that HYV benefits are not uniformly distributed among farmers, depending on the environmental conditions they face (Otsuka, 2000). Farmers in those marginal areas, such as highlands or less fertile soils, benefit less from HYVs. Thus, bringing the benefits of STVs to farmers in such areas can reduce the productivity gap arising from various production conditions. STVs, therefore, narrow the economic disparity among agriculture-dependent households in Nepal.

The remainder of this paper is organized as follows: Section 2 briefly explains our data collection procedures and empirical methods. Section 3 explains farmers' current variety choices, followed by the estimation results. Section 4 concludes the paper and discusses its policy implications.

2. Materials and methods

2.1. Location description

To explore the factors affecting rice variety adoption, we focused on two research sites with similar agro-environmental but different socioeconomic characteristics. Household surveys were conducted in October 2019 and October 2022 at the two study sites in the Chitwan district in central Terai: (1) the Bharatpur Metropolitan (Bharatpur) and (2) the Madi Rural Municipalities (Madi). Chitwan National Park (CNP) separates these two sites, Bharatpur in the north and Madi in the south of the CNP (Figure 1). Access to Madi from district's commercial center is limited and only possible through CNP. Access to CNP is not allowed at night and the road inside the park is not paved. Madi shares a border with India in the south but is surrounded by Chure. Madi is characterized as a less accessible area. Chitwan district faces high flood risks (Ministry of Environment, 2010). Thus, it is an appropriate district for this study – the adoption of a submergence-tolerant rice variety – as there is likely to be high demand for such varieties.

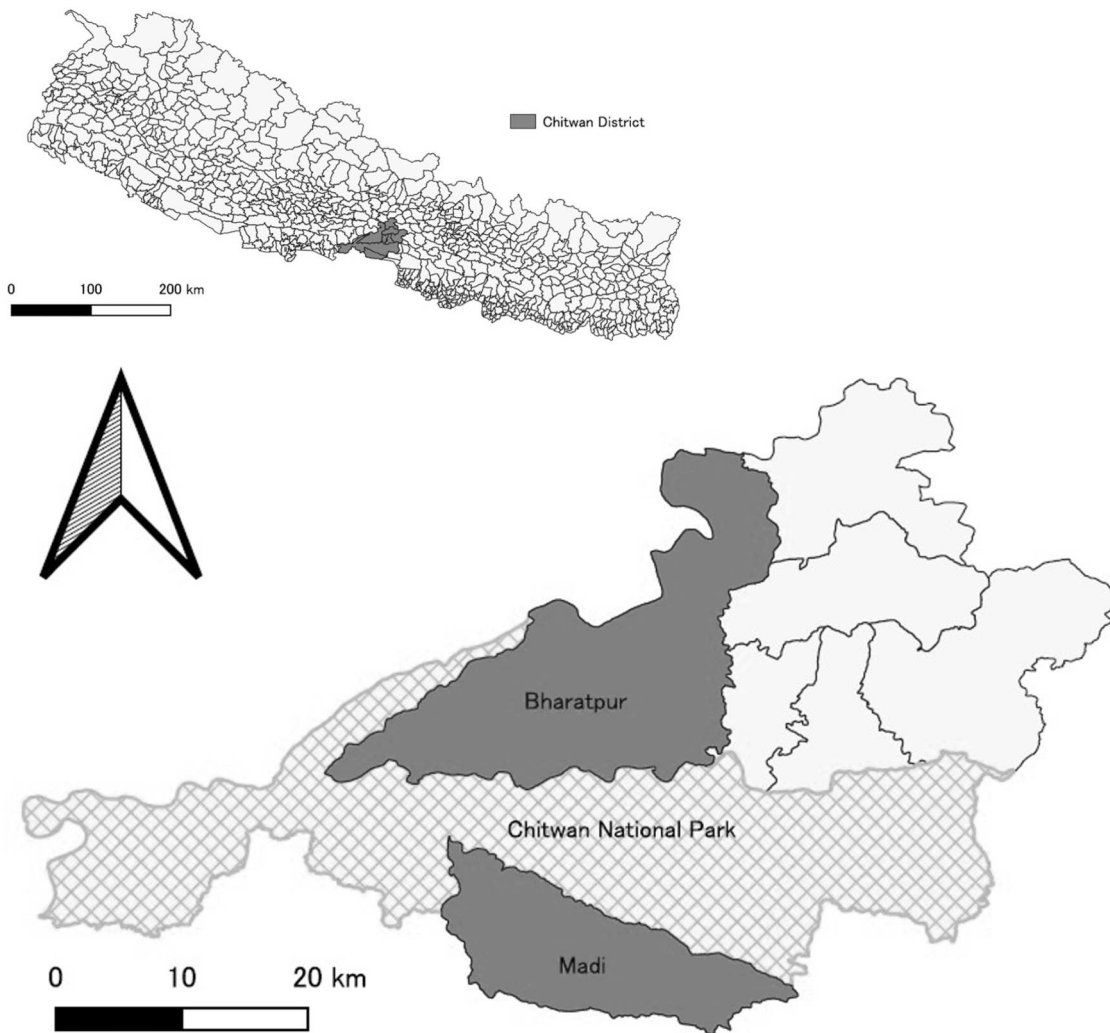


Figure 1. Description of study area.

2.2. Research design and procedure

2.2.1. Data collection

Information on farmers' variety choices are generally unavailable in secondary data, so that a farm household survey is inevitable. In October 2019, 380 households were selected based on stratified random sampling, then revisited in October 2022. Two hundred sixty-eight households were available for use in this two-period balanced panel data structure. Data drawn from the 2019 (2022) survey included agricultural activities in the 2018–2019 (2021–2022) fiscal years. Note that the fiscal year runs from 16th of July of the given calendar year to the 15th of July of the following calendar year. We visited sampled households and interviewed them with structured questionnaires. We used tablets that enabled us to recall the records of the previous (2019) survey in the 2022 survey and also to save time for interviews. The questions include those on household characteristics, agricultural production, and income sources. The survey was conducted by enumerators who are agricultural graduates from the study sites. Before implementing the survey, they were briefed about the research purpose, design, and implementation plan.

As rice variety choices in particular were not elicited in 2019 survey, they were collected in 2022 using the recall method. Ambiguity or bias may arise from the recall method and face-to-face interviews, some caution is needed in interpreting the results. Some farmers may poorly remember their variety choices in the earlier years, and lean rather toward desirable answers to the interview questions. However, we believe that it did not affect their responses enough to introduce much bias into the analysis, given that the recall period was relatively short.

Another limitation is the broader applicability of the results. Although Terai is the major rice growing region, the two sites selected from Chitwan district may not fully represent the entire region. Given the differences in rainfall variability and potential flood impact between eastern and western Terai, the study sites represent average rainfall and flood impact because Chitwan district is located in central Terai along the east–west continuum. The study sites could not represent the hilly regions of Nepal. The study sites are especially prone to flooding, and water stagnates for a longer time there than in the hill or mountain regions. Sample households were selected randomly but the representativeness of these households may be affected by microclimatic conditions such as flood frequency, which differs by region.

2.2.2. Surveyed variables

Definitions of the surveyed variables employed in the statistical analysis are given in Table 1. The upper panel – variety choice – contained the dependent variables, and the lower panel – household characteristics – the independent variables. The upper panel shows the farmers' variety choices between Sambha Mansuli SUB1 and Sabitri, the varieties most widely adopted. As rice variety choice is the most important variable in this survey, the response is open-based, not choice-based. Details are shown in Table 2. Household characteristics include such agricultural activities as area irrigated, distance to input markets, membership in cooperatives, access to extension services, and credit. Education years and total income are also included as household characteristics.

In our study, irrigation refers to 'year-round irrigation.' In the study areas, land is classified into two types: *Khet* and *Bari*. *Khet* refers to irrigated land or a paddy field. Most farmers plant rice in *Khet* land and use such irrigation systems as pump sets, a river canal, or even seasonal rainfall. As a result, water access can be seasonal, but we used any presence of irrigation systems as a proxy for water access.

2.3. Statistical analysis

We employ the pooled binomial logit model (PBLM) to investigate household characteristics affecting the likelihood of a given variety choice. The binominal logit approach is used to model technology adoption, especially the factors affecting the probabilities of a particular adoption (Mariano et al., 2012; Paudel & Matsuoaka, 2008). Pooled panel data were used to estimate changing factor impacts over time. Our analytical model then takes the form:

$$\Pr(Y_{l,i} = 1 | X_{1i}, \dots, X_{ki}) = \frac{\exp(\beta_0 + \sum^k \beta_k X_{ki})}{1 + \exp(\beta_0 + \sum^k \beta_k X_{ki})}, \quad (1)$$

$l = \text{Sambha Mansuli SUB1, Sabitri}$

where $Y_{l,i}$ is a dummy variable taking the value of 1 if household i cultivates variety l , otherwise 0; and X_{ik} representing the k^{th} characteristics of the i^{th} household (Table 1). β_0 and β_k are the parameters to be estimated. The two variety-choice equations, respectively for Sambha Mansuli SUB1 and Sabitri, are estimated independently.

Table 1. Definitions of variables.

Variables	Definition
Variety choice	
<i>sambha_choice</i>	Dummy taking 1 if Sambha Mansuli SUB1 is grown
<i>sabitri_choice</i>	Dummy taking 1 if Sabitri is grown
Household characteristics	
<i>area_irrigated</i> ^{*1}	Area with irrigation facilities available year-round (Kattha)
<i>input_market_distance</i>	Distance from nearest input market (Km)
<i>total_income</i> ^{*2}	Total income (NRs)
<i>years_edu</i>	Education years of household head (years)
<i>coop_member</i>	Dummy taking 1 if at least one member belongs to an agricultural cooperative
<i>extension_access</i>	Dummy taking 1 if reports easy access to agricultural extension services
<i>credit_access</i>	Dummy taking 1 if credit is available

Note 1: The variable *area_irrigated* is shown in Kattha, which is the local unit for agricultural land. 1 Kattha = 0.034 ha.

Note 2: Due to the limitation of data, we could not subtract the cost for income (i.e. in the case of agricultural income or own business, the number should rather be 'gross sale').

Table 2. Farmers' variety choices by survey year.

Variety combination			2019		2022	
Sambha Mansuli SUB1	Sabitri	Others	Number of farmers	Share in total (%)	Number of farmers	Share in total (%)
●			64	23.9	67	25.0
	●		92	34.3	80	29.9
●	●		10	3.7	13	4.9
●	●	●	3	1.1	4	1.5
●		●	24	9.0	23	8.6
	●	●	9	3.4	15	5.6
		●	61	22.8	64	23.9
		none	5	1.9	2	0.7
Total			268	100.0	268	100.0
Single variety only			217	81.0	211	78.7
Sambha Mansuli SUB1 or Sabitri			202	75.3	202 ⁽¹⁾	75.3
Sambha Mansuli (single or combined)			101	37.7	107 ⁽²⁾	39.9
Sabitri (single or combined)			114	42.5	112 ⁽³⁾	41.8

Note: Number of households of (1), (2), and (3) are further elaborated in Table 4.

The marginal effect of each variable on the likelihood of variety choice is given by the equation:

$$\frac{\partial \Pr(Y_{li} = 1 | X_{1i}, \dots, X_{ki})}{\partial X_{ki}} = \frac{\exp(\sum^k \hat{\beta}_k X_{ki})}{(1 + \exp(\sum^k \hat{\beta}_k X_{ki}))^2} \hat{\beta}_k \quad (2)$$

Eq. (1) is, for each of the two varieties, estimated using the maximum likelihood estimator and Eq. (2) calculated for each explanatory variable. Three methods can be used to deal with panel data in a binomial logit model: the PBLM, the random-effect (RE) binomial logit model, and the fixed-effect (FE) binomial logit. Households with time-invariant independent variables are excluded from the FE model (Cameron & Trivedi, 2005). Most household variety choices included in our data were time-invariant, so the FE was unsuitable. If our data were longitudinal, we would not be able to reject the possibility of serial correlation. Serial correlation in the RE logit are not assumed in our two year data (Wooldridge, 2010). Thus we will provide the PB logit model estimates using cluster-robust standard errors along with a region dummy and other variables expected to influence farmers' varietal choice decision. The analysis was conducted using STATA/MP ver. 15.0.

3. Results and discussion

3.1. Descriptive statistics of two major rice varieties

The numbers and shares of farmers producing given rice variety combinations in Terai are presented in Table 2. The two most popular are Sambha Mansuli-SUB1 and Sabitri. Others include improved varieties such as Rhada-4 and traditional ones like Anadi. As is shown in the lower panel, more than half of the households surveyed produced only one rice variety. The remaining produced more than one rice variety in both years. Approximately three quarters of the sampled households (75.3%) chose in both years to produce either Sambha Mansuli-SUB1 or Sabitri, making them the two major rice varieties at our study sites.

Sambha Mansuli-SUB1 is an improved variety with a submergence tolerance gene (SUB1) developed by the IRRI (Ismail et al., 2013; Singh et al., 2009) and released in 2011 (MoAD, 2015). By contrast, the SUB1 gene has not been scientifically identified in Sabitri, but is known to have a semi-dwarf gene. Based on our survey, the average seed price in 2022 was approximately NRs 86.5/kg for Sambha Mansuli-SUB1 and NRs 65.2/kg for Sabitri. Though Sambha Mansuli-SUB1 seeds were costlier, its average yield was expected to be higher than Sabitri.

Sambha Mansuli-SUB1's diffusion process differs from the old HYVs. In Chitwan this new stress-tolerant variety is available, so that farmers can benefit from the outcome of this maintenance-research. But STV's adoption rate remains low, partly evidenced by the popularity among farmers of the old HYVs. For example, though the Sabitri variety was released in 1979, it remains popular in both Bharatpur and Madi. Replacement of varieties prone to environmental stress is urgently needed to both reduce climate risk and enhance productivity and income.

Table 3. Descriptive statistics (means) of variables by region and survey year.

Variables	2019		2022	
	Madi	Bharatpur	Madi	Bharatpur
Variety choice				
<i>sambha_choice</i>	0.119	0.606***	0.150	0.62***
<i>sabitri_choice</i>	0.524	0.338***	0.465	0.38
Household characteristics				
<i>area_irrigated</i>	7.28	12.90***	4.90	12.76***
<i>input_market_distance</i>	2.79	2.18***	2.62	1.80***
<i>total_income</i>	414,677	1,212,046***	314,916	812,254***
<i>years_edu</i>	6.20	7.20*	5.54	7.96***
<i>coop_member</i>	0.75	0.94***	0.87	0.76**
<i>extension_access</i>	0.29	0.22	0.46	0.21***
<i>credit_access</i>	0.92	0.95	0.77	0.72
obs.	126	142	127	141
	268		268	

Note: ***, **, and * indicate that the regional difference is significance at the 1%, 5%, and 10% levels, respectively.

3.2. Regional heterogeneity

Descriptive statistics of the variables defined in Table 1 are presented in Table 3. The upper panel of Table 3 shows the sample means of the variety-choice dummy variables. Choice differed between the two regions. In Madi in 2019, 11.9% of farmers chose Sambha Mansuli-SUB1, and 52.4% chose Sabitri. By contrast, 60.6% of farmers in Bharatpur chose Sambha Mansuli-SUB1 and 33.8% chose Sabitri. A similar trend is observed for 2022. Sabitri was the main variety in Madi and Sambha Mansuli-SUB1 the main one in Bharatpur. The two regions are located near each other, but their choices significantly differ.

Household characteristics defined in Table 1 are shown in the lower panel of Table 3. In both years, access to agricultural resources – area irrigated and input market distance – are statistically different at the 1% significance level. In Bharatpur for example, a household's mean irrigation was 12.9 (kattha, 1 kattha = 338.2m²) in 2019, while in Madi it was 7.28. In 2022 similarly, the distance to the resource or input markets in Bharatpur was about 0.82 (2.62–1.80) km shorter than in Madi. Total household income in Bharatpur was also several times higher than in Madi. Commercial opportunities are abundant in Bharatpur, while Madi farmers have comparatively little access to nonagricultural commercial activities. Consequently, Madi farmers have become more self-reliant in both production and consumption. They were also significantly less educated than those in Bharatpur. Cooperatives and extension services are the two main sources of agricultural information in the Chitwan district. The participation rate in a cooperative was high compared to that in extension and significantly differed between the two regions.

3.3. Yield gap, input, and rice varieties

Table 4 compares the two varieties' average yields at each survey site in 2022. Average yields are calculated on the basis of four alternative input-use patterns: no input ('none'), irrigation only, pesticides only, and both irrigation and pesticides. The impact of a given input use is computed as a 'compared to none.' Corresponding regional gaps are shown at the bottoms of the panels. Though a test of statistical differences would be desirable, the number of observations is fewer for such tests. Even if significant differences were obtained, they would not be robust, and a large enough sample should be considered in future research to have confidence in statistical differences.

Overall, Bharatpur had a higher average yield than Madi. Where there was no use of pesticide or irrigation, average yield of a variety is far greater in Bharatpur. The regional gap in this 'none' or no input use case is 835 kg/ha for Sambha Mansuli SUB1 and 1,297 kg/ha for Sabitri. This gap however declined when 'irrigation only' was employed (664 kg/ha for Sambha Mansuli SUB1 and 888 kg/ha for Sabitri). Among farmers with both inputs, there was no clear regional gap in either variety. Despite that the production environment is less favourable in Madi, the yield gap can be reduced with proper irrigation and pest management. This means there is potential rice yield improvement in Madi. But as shown in Table 4 third panel, only 24 farms (7 'irrigation only' + 17 'both') of the 78 farms (30.8%) have access to irrigation in Madi, whereas

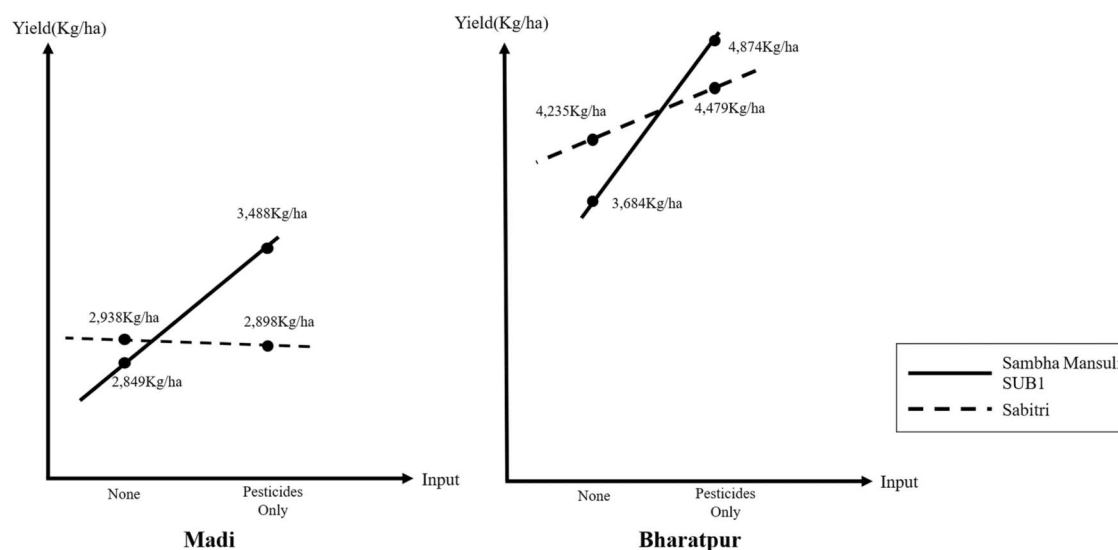
Table 4. Average yield of alternative agricultural practices (kg/ha) in 2022.

Region		None	Irrigation only	Pesticides only	Both	All
Panel 1: Sambha Mansuli SUB1 (n = 107) ⁽²⁾						
Madi	Average yield	2,849	4,319	3,488	4,492	3,345
	obs.	10	2	5	2	19
	Compared to None (%)	—	51.6	22.4	57.7	—
Bharatpur	Average yield	3,684	4,983	4,874	4,676	4,757
	obs.	7	40	9	32	88
	Compared to None (%)	—	35.2	32.3	26.9	—
Regional gap		−835.3	−664.0	−1,385.8	−183.9	—
Panel 2: Sabitri (n = 112) ⁽³⁾						
Madi	Average yield	2,938	3,593	2,898	4,113	3,287
	obs.	31	5	8	15	59
	Compared to None (%)	—	22.3	−1.4	40.0	—
Bharatpur	Average yield	4,235	4,481	4,479	4,090	4,319
	obs.	11	23	4	15	53
	Compared to None (%)	—	5.8	5.8	−3.4	—
Regional gap		−1,297.0	−887.8	−1,580.3	23	—
Panel 3: Both/either varieties (n = 202) ⁽¹⁾						
Madi	Average yield	2,916	3,800	3,125	4,157	3,301
	obs.	41	7	13	17	78
	share (%)	52.6	9.0	16.7	21.8	100
Bharatpur	Average yield	4,012	4,872	4,779	4,560	4,644
	obs.	17	55	12	40	124
	share (%)	13.7	44.4	9.7	32.3	100

Note: Households producing either Sambha Mansuli SUB1 or Sabitri were used for this analysis. Since 17 households produce both varieties (13 producing both and 4 producing these two or others), the total number of households from panel 1 and 2 cannot equal that in panel 3. Details of the sample size in three panels, indicated (1), (2), and (3), are found in Table 2.

76.6% (55 ‘irrigation only’ + 40 ‘both’ of the 124 farms) of those have such access in Bharatpur. Given that rainfed farmers preferred Sabitri (Yadaw et al., 2013), irrigation availability may be one of the main reasons farmers do not adopt Sambha Mansuli-SUB1 in Madi.

It is also known that Sabitri is less susceptible to pests than Sambha Mansuli-SUB1 (Regmi et al., 2017) and the additional yield from pesticide use (‘pesticides only’ in Table 4) differed here by variety (Figure 2). Pesticides had little effect on the average yield of Sabitri (dotted lines), but there is a notable effect on Sambha Mansuli-SUB1 (solid lines). When farmers used pesticides, the average yield of Sambha exceeded that of Sabitri in Madi. In other words, given the relatively low input access in Madi, the average yield of Sabitri (2,938 kg/ha) with ‘none’ was greater than that of Sambha Mansuli-SUB1 (2,849 kg/ha). But when farmers had access to pesticides the order of yield was reversed. The average yield of ‘pesticide only’ is 2,898 kg/

**Figure 2.** Impact of pesticide use on two rice varieties at the two research sites.

ha with Sabitri (−1.4%), but it increased to 3,488 with Sambha Mansuli-SUB1 in Madi (22.4%). A similar trend was observed in Bharatpur. Farmer's choice of Sabitri over Sambha is rational when access to pesticides is poor. Sabitri is a reasonable choice in limited input-access situation; but yield improvement is very high when switching to Sambha Mansuli-SUB1, aided by pesticides. This is consistent with Upadhyaya and Thapa (1994), who found that input access restrictions significantly reduced potential yield gains. When unaided by pesticides, Sambha Mansuli-SUB1's potential yield is constraint. These yield comparisons thus indicate that irrigation and pesticide access are key limiting factors, deterring farmers from shifting varieties to STV in the Chitwan district.

3.4. Estimated parameters and marginal effects

(1) Regional difference

Table 5 shows the PB logit model estimates for the Sambha Mansuli-SUB1 and Sabitri variety choices. The regional dummy shows a significantly negative effect on the Sambha choice. The marginal effect shows Madi farmers are less likely to adopt Sambha (58.6%). Sambha's adoption rate in Madi is much lower than that in Bharatpur.

As shown above, inputs access is comparatively weak in Madi, which is further from the Chitwan district's market center (e.g. Narayangadh) and on the southern side of the CNP, making access to the market center less convenient and more time-consuming. The regional dummy variables capture Madi's regional disadvantage in this regard. Poor access to input markets may be deterring Madi farmers from adopting Sambha, which requires proper pest management. Pesticide price can be relatively high in Madi because of its greater transportation costs from the market center. Given this relatively high pesticide cost, it is reasonable for Madi farmers to choose a variety requiring less input, i. e., Sabitri. This implies that future policy options to increase agricultural productivity in Madi should consider investing in more efficient delivery of, and easier access to, agricultural inputs. Historically, Madi has been a remote area with poorer access to the city center. Coexistence of wildlife and human activity is another issue in Madi because of its adjacency to the CNP.

Table 5. Results of PBLM estimation.

Variables	Sambha Mansuli Sub1		Sabitri	
	Coef.	Marginal effect	Coef.	Marginal effect
d_2022	0.258	0.044	−0.234	−0.052
d_Madi	−3.450***	−0.586***	0.612	0.136
area_irrigated	0.020*	0.003*	0.016*	0.004*
· d_Madi	−0.009	−0.002	−0.029	−0.006
· d_Madi · d_2022	−0.010	−0.002	0.035	0.008
total_income	0.002	0.0003	−0.002**	−0.001**
· d_Madi	0.002	0.0003	0.006	0.001
· d_Madi · d_2022	−0.011	−0.002	−0.008	−0.002
credit_access	0.662*	0.113*	−0.834**	−0.186**
· d_Madi	0.985	0.167	−0.778	−0.173
· d_Madi · d_2022	−1.759	−0.299	1.303*	0.290*
input_market_distance	0.002	0.000	−0.037	−0.008
· d_Madi	−0.132	−0.022	−0.118	−0.026
· d_Madi · d_2022	0.019	0.003	0.236	0.053
years_edu	0.020	0.003	−0.006	−0.0012
· d_Madi	0.159*	0.027*	0.042	0.009
· d_Madi · d_2022	−0.122	−0.021	−0.058	−0.013
coop_member	0.014	0.002	−0.520	−0.116
· d_Madi	−1.106	−0.188	1.067*	0.238*
· d_Madi · d_2022	3.831***	0.651***	−2.084***	−0.465***
extension_access	−0.015	−0.0026	0.111	0.025
· d_Madi	0.510	0.087	−0.205	−0.046
· d_Madi · d_2022	−0.697	−0.118	0.701	0.156
_cons	−0.746		0.738	
Number of obs	536		536	
Log pseudo likelihood	−271.53		−340.80	
Pseudo R ²	0.24		0.07	

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Effective implementation of a buffer zone policy, aiming to improve the living conditions of neighbouring communities in the CNP, is an option for enhancing Madi's social and economic infrastructure (Chitwan National Park Office, 2015; den Braber et al., 2018). It is crucial to implement effective policies to help Madi farmers adopt new STVs, and become less vulnerable to climate change. The income gap among farmers can be reduced in that way.

(2) Irrigation

Year-round irrigation has, as shown in Table 4, a significant positive effect on both varieties, indicating that improved water access would facilitate the choice of Sambha and Sabitri. Irrigating Sambha raises average yield by 51.6% (relative to 'none') in Madi and by 35.2% in Bharatpur. A similar pattern is found in Sabitri. When farmers have access to irrigation, both varieties begin to perform better, the rate of improvement in each being greater in Madi. As is shown in Table 5, the marginal effect of year-round irrigation lifted the probability of choosing a variety by 0.3% with Sambha and by 0.4% with Sabitri. Because, as indicated above, irrigation is presently less available in Madi, investment in irrigation facilities would be highly recommended.

(3) Income and credit access

Total household income significantly and negatively affects the choice of Sabitri but has no significant effect on the use of Sambha variety (Table 5). The marginal effect of credit access is also unfavourable to Sabitri, encouraging the use of Sambha. That is, farmers are likely to shift from Sabitri to Sambha when liquidity constraints are relaxed (Pandey, 2022). It is worth noting that while Sabitri is generally less productive of the two, it is also less susceptible to pests. If credit access were to be improved then, farmers could control pests while shifting to the more productive Sambha. This, however, was not the case in Madi in 2022, as indicated by our result for Sabitri ($credit_access \cdot d_Madi \cdot d_2022 = 0.29$).

Some farmers have reported that they faced a severe lack of agricultural inputs during and after the COVID-19 pandemic. Thus, even if they would had access to more credit, some would have had difficulty managing their inputs and so reverted to Sabitri in 2022. Emerick et al. (2016) indicate the SUB1 variety generally incentivizes agricultural investment, a matter of crowd-in, by boosting either land or fertilizer use. And our results do suggest that when financial pressure is relaxed, farmers will be more likely to select the SUB1 variety. Adopting SUB1 in turn further encourages input use on – and investment in – rice production.

(4) Education

In Madi, education also has significant implications in selecting Sambha, where one more year of education boosts Sambha use by 2.7%. Other studies have also encountered a positive effect of education (Ghimire et al., 2015; Ghimire & Huang, 2016; Yamano et al., 2015). Education is especially important for adopting STVs (Meher et al., 2022; Vergara et al., 2023). The higher the education, generally the better is the understanding of the functions of submergence tolerance, and so of the benefits of SUB1. It is particularly important when such traits are less obvious and only stochastically available.

We also find education to be important for adopting STV's. Our result for years of education in Madi ($year_s_edu \cdot d_Madi = 0.027$) suggests the marginal influence of another year of education is greater in Madi than in Bharatpur. This difference is due to the level of education typically attained in the two regions, with average of 6.2 years of schooling in Madi and 7.2 in Bharatpur. This one-year difference matches farmers' choices between primary and secondary education. Improving educational opportunities, especially for secondary education, is important for improving Madi's agricultural resilience.

(5) Cooperatives and extension

Cooperative membership is another variable significantly affecting varietal choice. By putting oneself into a position of improving one's local knowledge, becoming a cooperative member in Madi boosts the probability of selecting Sabitri ($coop_member \cdot d_Madi = 0.238$), especially when input access had initially been

weak. Interestingly, this impact was reversed in 2022, the interaction between cooperative membership and the Madi dummy at that point becoming negative at -0.465 . The rate of *affiliation* with a cooperative increased (Table 3), so that by 2022 the marginal effect of being a member of a cooperative began shifting farmers' variety choices away from Sabitri.

Local peer learning may also be an effective tool for influencing varietal adoption (Mehar et al., 2022; Yamano et al., 2018). Coop membership offers other services than just learning new farming methods (Bilal & Jaghdani, 2024). The primary reasons for joining agricultural cooperatives are (i) availability of credit services, (ii) access to inputs¹, (iii) technical training, and (iv) social bonding (Dhakal & Mueser, 2023). Consequently, cooperative membership improves *both* learning opportunities and supplies, and our results indicate that cooperatives are not just information hubs in Chitwan but an additional input *resource* as well. Promoting peer support through a coop thus can encourage adoption of better rice varieties.

(6) General discussion

Our findings on farmer varietal choices indicate that local contexts play an important role. Of the two dominant rice varieties considered here – Sambha Mansuli SUB1 (STV) and Sabitri (HYV) – the latter had the higher yield under resource-limited conditions. This is because Sabitri is less susceptible to pests than Sambha Mansuli-SUB1 is, and farmers in these conditions therefore will likely choose Sabitri. Yet the yield gap reverses once farmers begin using inputs in a more appropriate way. In other words, it is optimal for farmers to choose Sabitri particularly when they have limited access to inputs such as irrigation and pesticides.

Maintenance research, that aims to improve a variety's stress tolerance, is an important means of coping with changing climatic conditions. But adoption of the new STV's has been slow, and a farmer's adoption decision will vary according to the local agro-climatic and socio-economic situation. In certain areas pest tolerance will be preferred, so shifts to new varieties susceptible to pests will be avoided. By contrast other areas are benefiting from a new variety's stress tolerance. Farmers in Chitwan district face a tradeoff between two alternative versions of varietal benefit: either pest tolerance from Sabitri or flood tolerance from Sambha. Depending on the socio-economic setting and microclimatic conditions such as flood frequency, they would choose the more suitable variety for their own fields. This is especially so in the resource-poor areas of Nepal (Sthapit et al., 2008). Once input availability changes, re-optimization of varietal choice on a family farm is to be expected. Maintenance research is a promising long-term effort to continue to improve a varietal's property. Because the effects of climate change have been increasingly severe, international attention to areas less capable of breeding research is urgently needed.

4. Conclusion

Given the increasing vulnerability under climate change, agricultural production especially in developing countries like Nepal is in need of greater resilience. Stress tolerant varieties is one adaptation strategy, but the diffusion of such variety remains slow. We have investigated farmers' adoption decisions among stress tolerant varieties, in particular submergence-tolerant rice varieties. Farm variety choices seem unreasonable, but understanding them in their local contexts can help solve this puzzle.

Results of yield comparisons show that poor access to pesticides is one reason farmers choose an old high-yielding variety, which is tolerant to pests. Choosing old HYVs is reasonable for resource-poor farmers. Policy options improving the input access are worth considering. Peer support can also encourage farmers to turn to new varieties. Policy makers can encourage farmer networks and disseminate technical knowledge of stress-tolerant varieties. These measures can reduce regional gaps in varietal choice as well as economic situation, helping develop a resilience to increasing vulnerability to climate variation.

Bringing the Green Revolution to marginal areas by way of varietal maintenance research can reduce social disparities and lead to a more sustainable agriculture. Research that would improve farmer resilience to climate change is urgently required not only in Nepal and South Asia but in Africa, which too confronts serious climate change problems.

Note

1. According to Dhakal and Mueser (2023), farmers do not legally have to be a member of a cooperative in order to purchase subsidized inputs. They note however social obligations are also involved. Once farmers purchase inputs from a cooperative, they are expected to become its member. In this local context, membership in a coop or farmer group is likely to be related to agricultural input access.

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